

SEMINAR REPORT ON

LONG-RANGE, OFF-GRID EMERGENCY ALERT MESSAGING SYSTEM USING LORA TECHNOLOGY

SUBMITTED BY
Abhijeet Arun Dalvi
Atharv Ranveer Pawar
Bhushan Sandeep Todkar

UNDER THE GUIDANCE OF

Prof. S. L. Tatuskar

**Department of Electronics and
Telecommunication Engineering
Program: ECE**



**Progressive Education Society's
Modern College of Engineering
Shivajinagar, Pune-05**



**SAVITRIBAI PHULE PUNE UNIVERSITY
ACADEMIC YEAR: 2025-26**



CERTIFICATE

This is to certify that

1. Abhijeet Arun Dalvi
2. Atharv Ranveer Pawar
3. Bhushan Sandeep Todkar

Exam No: T400310955

Exam No:T400310986

Exam No:T400311003

of T.E. E&TC have successfully completed the Project titled 'LONG-RANGE, OFF-GRID EMERGENCY ALERT MESSAGING SYSTEM USING LORA TECHNOLOGY' during the academic year 2025-26 for the Course Mini Project. This report is submitted as per the requirement prescribed by Savitribai Phule Pune University.

Dr. Mrs. R. S. Kamathe
Vice Principal, H.O.D. (E&TC)

Prof. S. L.Tatuskar
Project Guide

DECLARATION

We declare that this project report titled “**Long-Range, Off-Grid Emergency Alert Messaging System Using LoRa Technology**” submitted in partial fulfilment of the degree of **BE ECE** is a record of original work carried out by us under the guidance of Prof. S. L. Tatuskar, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University.

We declare that this written submission represents the ideas in our own words. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in the submission.

- | | |
|---------------------------|-------------------|
| 1. Abhijeet Arun Dalvi | T400310955 |
| 2. Atharv Ranveer Pawar | T400310986 |
| 3. Bhushan Sandeep Todkar | <u>T400311003</u> |

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the development of our project, “**Long-Range, Off-Grid Emergency Alert Messaging System using LoRa Technology.**”

First and foremost, we extend our sincere thanks to our project guide, **Prof. S. L. Tatuskar**, for his continuous support, expert guidance, and valuable feedback throughout the project. His encouragement and insightful suggestions played a crucial role in overcoming technical challenges and enriching our learning experience.

We are deeply grateful to our **Head of Department, Dr. Mrs. Rupali Kamathe**, for her constant support and providing us with the opportunity to work on this innovative project. Their guidance has been instrumental in shaping our academic journey.

Our heartfelt appreciation goes to **Dr. Mrs. K.R. Joshi Ma’am, Principal, PES Modern College of Engineering**, for their motivation and support, which inspired us to strive for excellence.

We would also like to thank all the faculty members and lab assistants of the Department of Electronics and Telecommunication Engineering for providing the necessary resources and technical assistance during the implementation and testing phases.

We sincerely acknowledge the support of our college and department for providing us with a platform to explore our technical interests and apply our knowledge to practical challenges.

Finally, we would like to thank our families and friends for their unwavering encouragement, patience, and motivation. Their support has been the foundation of our dedication and hard work.

Abhij Arun Dalvi
Atharv Ranveer Pawar
Bhushan Sandeep Todkar

ABSTRACT

This project presents the design and implementation of a long-range off-grid emergency communication system using LoRa technology. The system is developed to provide reliable communication in remote, forest, rural, and disaster-affected areas where cellular networks and internet connectivity are unavailable or unreliable. The proposed model uses LoRa SX1278 modules operating at 433 MHz for low-power and long-distance wireless transmission of emergency messages. STM32 Blue Pill microcontrollers are used for processing and control of the communication system. Users can send predefined emergency alerts such as “HELP” and “SAFE” using push buttons, while the received messages are displayed on an OLED screen along with LED indications. The system also supports GPS integration using the NEO-6M module for transmitting location coordinates during emergencies. Since the system operates independently without requiring SIM cards or internet infrastructure, it is highly suitable for disaster management, trekking, rescue operations, and remote area communication with minimal power consumption.

TABLE OF CONTENT

DESCRIPTION	PAGE NUMBER
CERTIFICATE	ii
DECLARATION	iii
ACKNOWLEDGEMENT	iv
ABSTRACT	v
CHAPTER 1: INTRODUCTION	
1.1 Project Idea	1
1.2 Motivation	1
1.3 Objectives	2
1.4 Scope of Project	3
1.5 Overview of Project	3
CHAPTER 2: LITERATURE SURVEY	
2.1 Literature Review	4
2.2 Market Survey	6
2.3 Component Survey	7
2.4 Observations	7
2.5 Problem Statement	8
2.5.1 Aim and Objectives	8
2.5.2 Scope	9
2.6 Methodology	9
2.7 Outcome	10
2.8 Applications	10
CHAPTER 3: SPECIFICATIONS	
3.1 Electrical Specifications	11
3.2 Mechanical Specifications	11
CHAPTER 4: BLOCK DIAGRAM & DESCRIPTION	
4.1 Block Diagram	12
4.2 Description	13-14
CHAPTER 5: HARDWARE SYSTEM DESIGN	
5.1 Hardware Resources	15
5.2 Description of Hardware	15
5.3 Enclosure Design	16
CHAPTER 6: SOFTWARE SYSTEM DESIGN	
6.1 Software Resources	17
6.2 Algorithm	18
6.3 Flowchart	19

CHAPTER 7: RESULTS AND PERFORMANCE EVALUATION	20-24
CHAPTER 8: CONCLUSION AND FUTURE SCOPE	25-26
CHAPTER 9: BILL OF MATERIAL	27
REFERENCES	28

CHAPTER 1

INTRODUCTION

1.1 Project Idea

In recent years, reliable communication has become a critical requirement, especially in emergency situations such as natural disasters, remote area operations, and rescue missions. Conventional communication systems like cellular networks and internet-based services often fail in such scenarios due to infrastructure damage or lack of coverage. This creates a significant communication gap, making it difficult to send or receive crucial information during emergencies.

The primary idea of this project is to develop a long-range, off-grid communication system using LoRa technology that can function independently of existing communication infrastructure. The system is designed to transmit predefined emergency messages over long distances with minimal power consumption, ensuring reliable connectivity even in challenging environments.

1.2 Motivation

The motivation behind this project arises from the increasing need for dependable communication systems in disaster-prone and remote regions. Situations such as earthquakes, floods, forest operations, and mountainous expeditions often lack reliable network connectivity. In such cases, timely communication can play a vital role in saving lives.

Existing communication systems are either costly, power-intensive, or dependent on external infrastructure. This project aims to overcome these limitations by providing a low-cost, energy-efficient, and easy-to-deploy communication solution using LoRa technology.

1.3 Objectives

The main objectives of this project are:

- To design and implement a wireless communication system using LoRa technology.
- To enable long-range transmission of emergency messages without using cellular networks or internet.
- To develop a user-friendly interface for sending and receiving predefined alert messages.
- To minimize power consumption for extended operation using battery supply.
- To demonstrate reliable communication in non-line-of-sight and remote conditions.

1.4 Scope of Project

The scope of this project includes the design and development of a point-to-point wireless communication system using LoRa modules. The system focuses on transmitting short text-based emergency messages between two nodes over long distances.

The project is limited to a prototype implementation and does not include large-scale network deployment or mesh networking. However, the system can be further extended into a multi-node communication network for broader applications.

The proposed system has potential applications in disaster management, rural communication, military operations, and search and rescue missions where conventional communication systems are unreliable or unavailable.

1.5 Overview of Project

The proposed system consists of two main units: a transmitter and a receiver, both equipped with LoRa SX1278 modules for long-range wireless communication. The transmitter unit includes a microcontroller, input interface (such as push buttons or keypad), and a LoRa module to send predefined emergency messages. The receiver unit consists of a LoRa module interfaced with a microcontroller and an OLED display to receive and display the transmitted messages.

When a user selects an emergency message at the transmitter side, the microcontroller processes the input and transmits the message via the LoRa module. The receiver unit captures the signal, decodes the message, and displays it on the screen in real time.

The system is designed to operate on low power, making it suitable for battery-operated applications. Its ability to function without relying on cellular networks or internet connectivity makes it highly effective in remote and disaster-prone areas.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Review

[1] Q. Zhou, K. Zheng, L. Hou, J. Xing and R. Xu, "Design and Implementation of Open LoRa for IoT"

The authors presented the design and implementation of an open LoRa communication platform for Internet of Things (IoT) applications. The research focused on improving long-range wireless communication with low power consumption using LoRa technology. The paper explained the architecture, communication mechanism, and performance analysis of LoRa networks. The study demonstrated that LoRa provides reliable long-distance communication with reduced energy consumption, making it suitable for smart city, industrial, and remote monitoring applications. This work highlights the effectiveness of LoRa technology for low-power off-grid communication systems.

[2] S, Blesson. (2024). LoRa Based Emergency Communication System. International Scientific Journal of Engineering and Management.

This paper proposed a LoRa-based emergency communication system designed for disaster-prone and remote areas where cellular communication is unavailable. The system enabled transmission of emergency alerts over long distances using LoRa modules. The author emphasized low power consumption, portability, and independent operation without internet infrastructure. The research demonstrated that LoRa technology can be effectively used for emergency messaging applications and disaster communication. The proposed system provided inspiration for developing reliable off-grid communication systems using LoRa and embedded controllers.

[3] Darshani Mahakal, Manan Bhavsar, Aditya Mane, Prof. Mayuresh Bakshi, Dr. Shraddha Habbu, 2025, Cordless Communication system using LoRa Technology

The authors developed a cordless communication system using LoRa technology for long-distance wireless data transmission. The paper discussed the advantages of LoRa over conventional wireless technologies such as Wi-Fi and Bluetooth in terms of communication range and power efficiency. The proposed system demonstrated reliable transmission of messages between devices without requiring network infrastructure. The study concluded that LoRa is highly suitable for communication in rural and remote regions. The research supports the implementation of low-cost and energy-efficient emergency communication systems.

[4] Effective Communication System for Disaster Management Using LoRA , E. Senthilvel Murugan, M.Shunmuga Lakshmi and A. Ramya Lakshmi

This research focused on developing an effective disaster management communication system using LoRa technology. The system aimed to maintain communication during emergencies such as floods, earthquakes, and natural disasters where traditional communication systems may fail. The paper highlighted the long-range capability, low power consumption, and reliability of LoRa communication in disaster scenarios. The authors proposed a wireless communication model capable of transmitting alerts and emergency information efficiently in remote areas. The study demonstrated the importance of LoRa-based systems for disaster response and rescue operations.

2.2 Market Survey

Currently, most communication systems available in the market rely on cellular networks or satellite communication. Mobile phones are the most commonly used devices, but they become ineffective in areas with poor network coverage or during disasters when network infrastructure is damaged.

Satellite communication systems provide reliable connectivity but are expensive and not easily accessible for common users. Walkie-talkies offer short-range communication but are limited in range and scalability.

There is a growing demand for low-cost, long-range communication systems that can operate independently of existing infrastructure. LoRa-based devices are gaining popularity due to their affordability, low power consumption, and long communication range, making them suitable for emergency communication applications.

2.3 Component Survey

The major components used in this project include the LoRa SX1278 module, microcontroller, OLED display, and input interface.

The SX1278 LoRa module operates at 433 MHz frequency and supports long-range communication with low power consumption. It uses spread spectrum modulation techniques, which enhance signal reliability and resistance to interference.

Microcontrollers such as Arduino or STM32 are commonly used for processing and controlling the system. These controller interface with the LoRa module and manage data transmission and reception.

OLED displays are preferred for their low power consumption, compact size, and clear visibility. They are used to display received messages at the receiver end.

Input devices such as push buttons or keypads are used to select predefined emergency messages.

2.4 Observations from Literature and Market Survey

From the literature and market survey, it is observed that existing communication systems either depend heavily on infrastructure or are expensive and complex. While LoRa technology offers a promising solution, most implementations focus on data logging and IoT applications rather than emergency communication.

There is a lack of simple, portable, and cost-effective systems specifically designed for emergency alert messaging. Additionally, user-friendly interfaces and predefined messaging systems are not commonly implemented in existing solutions.

2.5 Problem Statement

In remote and disaster-affected areas, reliable communication becomes a major challenge due to the failure or absence of conventional communication infrastructure such as cellular networks and internet services. This creates difficulty in sending critical emergency information, leading to delays in response and rescue operations.

Therefore, there is a need to develop a communication system that can operate independently of existing infrastructure and provide reliable long-range communication during emergencies.

2.5.1 Aim and Objectives

Aim:

To design and implement a long-range, off-grid emergency alert messaging system using LoRa technology.

Objectives:

- To establish communication without relying on cellular networks or internet.
- To enable transmission of predefined emergency messages.
- To ensure low power consumption for extended operation.
- To achieve long-range communication using LoRa modules.

2.5.2 Scope

The scope of this project is limited to the design and implementation of a point-to-point communication system using LoRa modules. The system focuses on transmitting short emergency messages between two devices.

It does not include advanced networking features such as multi-node communication or real-time tracking. However, the system can be extended in the future for more complex applications.

2.6 Methodology

The system is developed using a transmitter-receiver model. The transmitter unit allows the user to select predefined emergency messages through input buttons. The selected message is processed by the microcontroller and transmitted using the LoRa module.

At the receiver end, the LoRa module receives the signal and passes it to the microcontroller, which decodes the message and displays it on the OLED .

The design focuses on simplicity and low power consumption.

2.7 Outcome

The expected outcome of the project is a functional prototype capable of transmitting emergency messages over long distances without relying on external communication infrastructure. The system demonstrates reliable communication, low power consumption, and ease of use, making it suitable for emergency applications.

2.8 Applications

- Disaster management and rescue operations
- Communication in remote and rural areas
- Military and defense applications
- Forest and wildlife monitoring
- Trekking and expedition safety systems

CHAPTER 3

SPECIFICATIONS

3.1 Electrical Specifications

The electrical specifications of the proposed system define the operating parameters of the hardware components used in the project.

- Microcontroller: STM32 Blue Pill (STM32F103C8T6)
- Operating Voltage: 3.3V
- LoRa Module: SX1278 (Operating Frequency: 433 MHz)
- Communication Range: Up to 5–10 km (depending on terrain and obstacles)
- Modulation Technique: Chirp Spread Spectrum (CSS)
- Data Rate: 0.3 kbps to 50 kbps
- Display: 128x64 OLED (I2C interface)
- Input Devices: Push buttons (GPIO based)
- **Communication Interfaces:**
 - SPI (LoRa module)
 - I2C (OLED display)
 - GPIO (input buttons)
- Power Supply: 3.7V Li-ion battery with voltage regulation to 3.3V
- Power Consumption: Low power, suitable for battery-operated applications
- Transmission Mode: Half-duplex communication

3.2 Mechanical Specifications

The mechanical specifications describe the physical and structural characteristics of the system. Enclosure Type: Compact handheld casing

- Dimensions: Approx. 10 cm × 8 cm × 4 cm (customizable)
- Weight: Less than 500 grams
- Material: Acrylic / ABS plastic enclosure
- Portability: Designed for easy handling and field deployment
- Mounting: Handheld or fixed mounting possible
- Power Source: Rechargeable Li-ion battery

CHAPTER 4

BLOCK DIAGRAM & DESCRIPTION

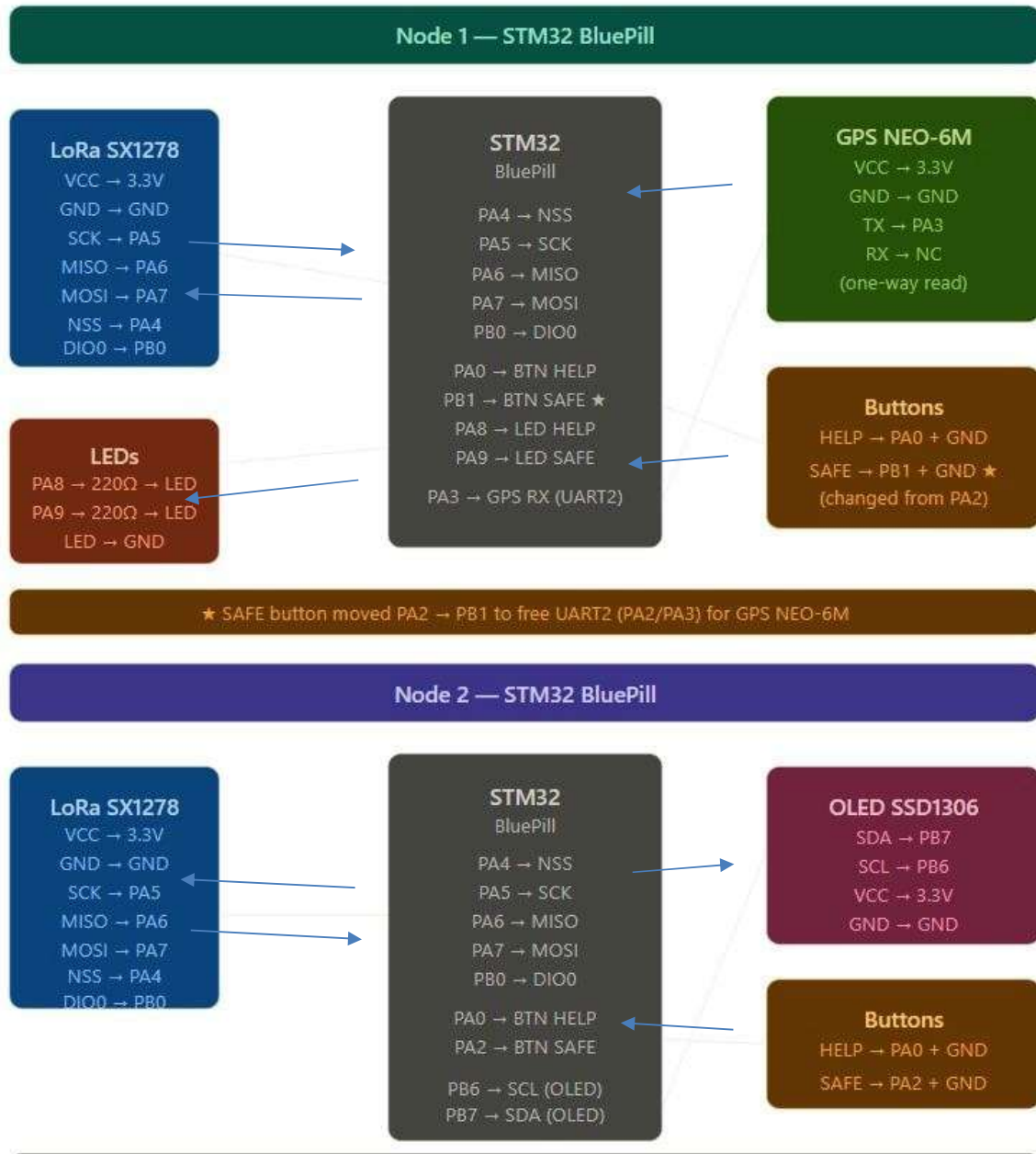


Fig. No. 1- Block Diagram

4.1 System Overview

The proposed system consists of two nodes built using STM32 Blue Pill microcontrollers, communicating via LoRa SX1278 modules. Node 1 acts as the transmitter, while Node 2 acts as the receiver. The system enables long-range wireless transmission of emergency messages without relying on cellular networks.

4.2 Transmitter Node (Node 1)

The transmitter node includes the following components:

- STM32 Blue Pill Microcontroller: Acts as the central processing unit and controls all peripherals.
- LoRa SX1278 Module: Connected via SPI interface (PA4–PA7), used for long-range wireless transmission.
- Push Buttons: Connected to GPIO pins (PA0 and PB1) for selecting emergency messages such as “HELP” and “SAFE.”
- LEDs: Used as status indicators to show message transmission.
- GPS NEO-6M Module: Connected via UART (PA3), used to obtain location data (optional feature).

Working:

When a button is pressed, the STM32 reads the input and sends a predefined message via the LoRa module. If GPS is enabled, location data can also be included in the transmitted message.

4.3 Receiver Node (Node 2)

The receiver node consists of:

4.3.1 STM32 Blue Pill Microcontroller: Processes received data.

4.3.2 LoRa SX1278 Module: Receives transmitted messages via SPI interface.

4.3.3 OLED Display (SSD1306): Connected via I2C (PB6, PB7), used to display received messages.

4.3.4 Push Buttons: Used for basic input or testing purposes.

Working:

The LoRa module continuously listens for incoming signals. Upon receiving a message, the STM32 processes it and displays the corresponding alert on the OLED screen.

4.4 Communication Interfaces

4.4.1 SPI Protocol: Used for communication between STM32 and LoRa module.

4.4.2 I2C Protocol: Used for interfacing OLED display with STM32.

4.4.3 UART Protocol: Used for GPS module communication.

4.4.4 GPIO: Used for buttons and LEDs.

4.5 Working Principle

The system operates on a point-to-point communication model. When a user presses a button on the transmitter node, the STM32 microcontroller encodes the selected emergency message and transmits it using the LoRa SX1278 module.

The receiver node captures the signal, decodes the message, and displays it on the OLED screen. The use of LoRa technology ensures long-range communication with low power consumption, making the system suitable for off-grid emergency applications.

CHAPTER 5

HARDWARE SYSTEM DESIGN

5.1 Hardware Resources:

Table 1: Hardware Requirements

Sr. No	Component	Minimum Requirement	Justification
1	STM32 Blue Pill	STM32F103C8T6, 72 MHz, 3.3V	Main controller
2	LoRa SX1278 Module	433 MHz, SPI interface	Long-range communication
3	OLED Display (SSD1306)	128x64, I2C interface	Display received messages
4	GPS Module (NEO-6M)	UART interface	Location tracking
5	Push Buttons	GPIO input	Message selection
6	LEDs	220 Ω resistor	Status indication
7	Li-ion Battery	3.7V	Power supply
8	Voltage Regulator	3.3V output	Stable power supply
9	Connecting Wires	Standard jumper wires	Interconnections

5.2 Description of Hardware used:

STM32 Blue Pill:

The STM32F103C8T6 microcontroller is used as the core processing unit. It operates at 72 MHz and supports multiple communication protocols such as SPI, I2C, and UART, making it suitable for interfacing with LoRa modules, OLED display, and GPS module.

LoRa SX1278 Module:

The SX1278 module operates at 433 MHz and uses Chirp Spread Spectrum modulation. It enables long-range communication with low power consumption and is interfaced with the STM32 using SPI protocol.

OLED Display (SSD1306):

The OLED display is used to show received messages. It operates on I2C communication, requiring only two data lines (SDA and SCL), and provides clear visibility with low power usage.

GPS Module (NEO-6M):

The GPS module provides real-time location data. It communicates with the STM32 via UART and can be used to include location information in emergency messages.

Push Buttons:

Push buttons are used as input devices to select predefined emergency messages. They are connected to GPIO pins and provide a simple user interface.

LED Indicators:

LEDs are used to indicate system status such as message transmission or power status. Current limiting resistors (220Ω) are used to protect the LEDs.

Power Supply:

The system is powered using a 3.7V Li-ion battery. A voltage regulator ensures a stable 3.3V supply required for STM32 and LoRa module operation.

5.3 Enclosure design

The system is designed to be compact and portable. A lightweight enclosure made of acrylic or plastic is used to house all components securely. The enclosure includes openings for buttons, display and charging port.

CHAPTER 6

SOFTWARE SYSTEM DESIGN

6.1 Software Resources

The software tools and platforms used for the development of this project are as follows:

- Operating System: Windows 10 / 11
- IDE: STM32CubeIDE/Arduino IDE
- Programming Language: Embedded C
- Debugging Tool: ST-Link Debugger

STM32CubeIDE/Arduino IDE is used for writing, compiling, and uploading the code to the STM32 Blue Pill microcontroller. It provides an integrated environment for peripheral configuration, code development, and debugging.

6.2 Algorithm

The system follows a simple transmitter-receiver algorithm for communication.

Transmitter Algorithm:

1. Start
2. Initialize GPIO, SPI, and UART peripherals
3. Initialize LoRa module
4. Continuously monitor input buttons
5. If button is pressed:
 - Identify the selected message
 - (Optional) Read GPS data
 - Encode the message
 - Transmit via LoRa module
6. Repeat

Receiver Algorithm:

1. Start
2. Initialize GPIO, SPI, and I2C peripherals
3. Initialize LoRa module
4. Continuously listen for incoming signals
5. If message received:
 - Decode the message
 - Display on OLED screen
6. Repeat

6.3 Flowchart

The flow of the system operation is illustrated using the following steps:

Transmitter Flow:

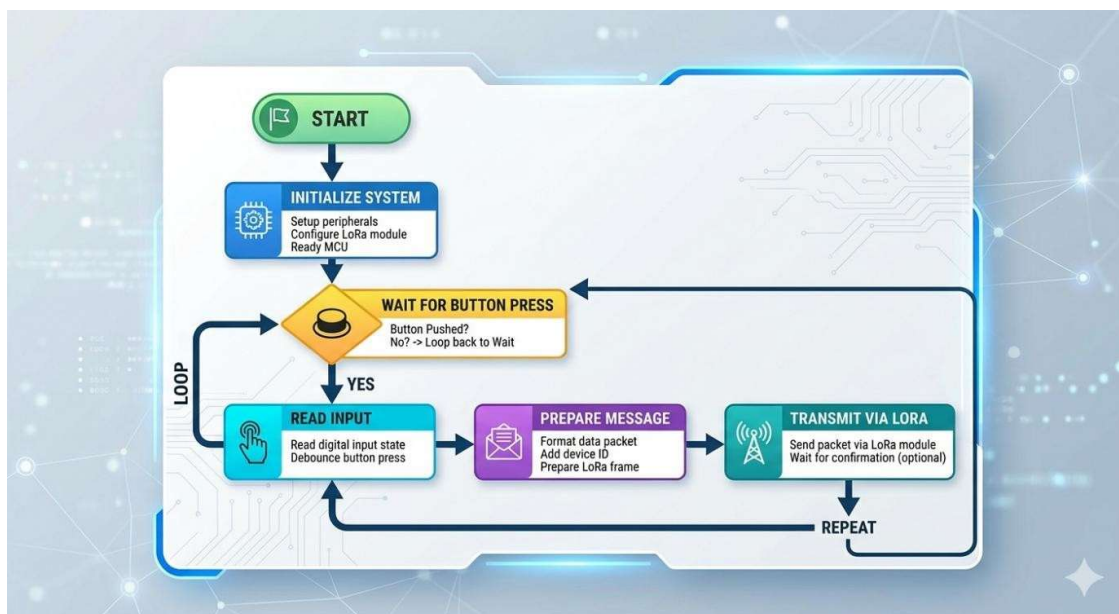


Fig. 2(a)- Transmission Flow Chart

Receiver Flow:

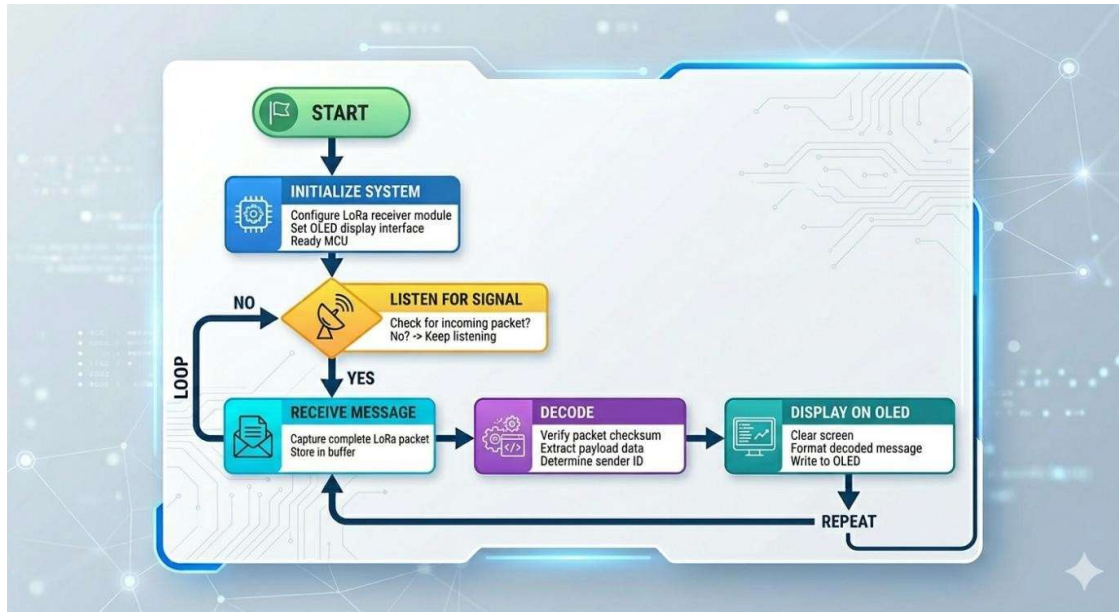


Fig. 2(b)- Receiver Flow Chart

CHAPTER 7

RESULTS AND PERFORMANCE EVALUATION

8.1 Photo of the Project with Enclosure:



Fig. 3- Node 2 Ready for Reception

8.2 Photo of the Project without Enclosure all the Hardware Modules:

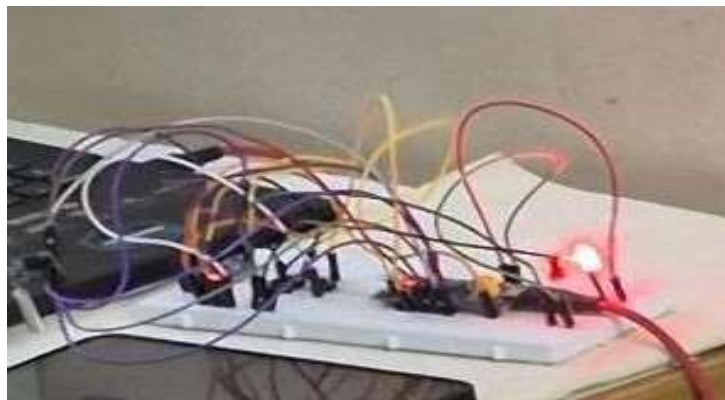


Fig. 4- LED glow indicating HELP message on Node 2

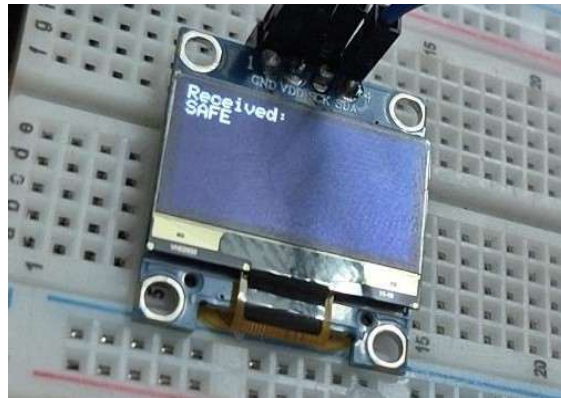


Fig. 5- SAFE Message Received on Node1

8.3 Operating Procedure of the Project:

The operation of the system is simple and user-friendly. The following steps describe the working procedure of the project:

1. Power ON both the transmitter node (Node 1) and receiver node (Node 2) using the battery supply.
2. Ensure that both LoRa modules are properly initialized and connected.
3. On the transmitter side, the user selects the desired emergency message by pressing the corresponding push button (e.g., HELP or SAFE).
4. The STM32 microcontroller reads the button input and prepares the predefined message for transmission.
5. (Optional) If GPS is enabled, the system reads the location data from the GPS module and appends it to the message.
6. The message is transmitted wirelessly using the LoRa SX1278 module.
7. At the receiver side, the LoRa module continuously listens for incoming signals.
8. Upon receiving the signal, the STM32 microcontroller decodes the message.
9. The decoded message is displayed on the OLED screen for the user.
10. The system remains in continuous operation, allowing multiple messages to be transmitted and received.

8.4 Necessary Precautions:

The following precautions should be taken to ensure proper functioning & safety of the system:

- Ensure that all connections are properly secured before powering the system.
- Use a regulated 3.3V power supply to avoid damage to STM32 and LoRa modules.
- Avoid reverse polarity connections, especially for battery terminals.
- Do not operate the system in extremely wet or humid conditions.

- Ensure proper antenna connection to the LoRa module for optimal range.
- Avoid short circuits while handling the circuit.
- Handle electronic components carefully to prevent static damage.
- Keep the system within recommended operating voltage limits.
- Ensure that GPS module has clear sky visibility for accurate location data.
- Switch off the system when not in use to conserve battery power.

8.5 Results:

The developed system was successfully tested for long-range wireless communication using LoRa technology. The transmitter and receiver nodes were able to establish reliable communication without the use of cellular networks or internet connectivity.

Emergency messages such as “HELP” and “SAFE” were transmitted successfully from the transmitter node and displayed accurately on the OLED screen at the receiver node. The system demonstrated stable communication over a considerable distance under non-line-of-sight conditions.

The power consumption of the system was observed to be low, making it suitable for battery-operated applications. The optional GPS module was also able to provide location data, which can be integrated into the transmitted message.

The results confirm that the proposed system is effective for emergency communication in remote and off-grid environments. The system meets the objectives of long-range communication, low power consumption, and ease of use.

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The proposed long-range off-grid emergency communication system successfully demonstrates reliable wireless communication using LoRa technology without requiring internet or cellular networks. The system effectively transmits emergency messages and GPS coordinates over long distances with low power consumption. Integration of STM32 microcontrollers, LoRa SX1278 modules, OLED display, LEDs, and GPS module provides a portable, cost-effective, and user-friendly solution for remote communication. The project is highly suitable for disaster management, trekking, rescue operations, and rural or forest environments where conventional communication systems are unavailable. Overall, the system proves to be an efficient and dependable solution for emergency and off-grid communication applications.

9.2 Future Scope

Although the current system performs effectively, there is significant scope for further improvement and enhancement.

- The system can be extended to support multi-node communication, enabling the creation of a mesh network.
- Integration with mobile applications or web interfaces can improve usability and monitoring.
- Advanced features such as real-time GPS tracking and mapping can be implemented.
- Encryption techniques can be added to enhance data security.
- Solar charging systems can be incorporated for sustainable and long-term operation.
- Voice communication or multimedia transmission can be explored in future versions.

These enhancements can make the system more robust, scalable, and suitable for real- world large-scale deployment in disaster management and remote communication scenarios.

CHAPTER 10

BILL OF MATERIAL

Sr. No.	Component Name	Component Specification	Quantity	Rate (Rs)	Amount (Rs)
1	STM32 Blue Pill	STM32F103C8T6 Microcontroller	2	250	500
2	LoRa SX1278 Module	433 MHz, SPI Interface	2	450	900
3	LoRa SX1278 Module	128x64 SSD1306 (I2C)	1	200	200
4	GPS Module	NEO-6M	1	300	300
5	Push Buttons	Tactile Switch	4	10	40
6	LEDs	5mm LED	4	5	20
7	Resistors	220 Ω , 1/4W	4	2	8
8	Li-ion Battery	3.7V Rechargeable	2	150	300
9	Voltage Regulator	AMS1117-3.3V	2	20	40
10	Connecting Wires	Jumper Wires	1 Set	100	100
11	PCB / Breadboard	General Purpose	1	150	150

REFERENCES

Q. Zhou, K. Zheng, L. Hou, J. Xing and R. Xu, "Design and Implementation of Open LoRa for IoT", IEEE Access, vol. 7, pp. 100649-100657, 2019 [1]

S, Blesson. (2024). LoRa Based Emergency Communication System. International Scientific Journal of Engineering and Management. 03. 1-9. 10.55041/ISJEM01628.[2]

Darshani Mahakal, Manan Bhavsar, Aditya Mane, Prof. Mayuresh Bakshi, Dr. Shraddha Habbu, 2025, Cordless Communication system using LoRa Technology, INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) Volume 14, Issue 11 , November – 2025, 10.17577/IJERTV14IS110335 [3]

Effective Communication System for Disaster Management Using LoRA , E. Senthilvel Murugan, M.Shunmuga Lakshmi and A. Ramya Lakshmi [4]

Pasolini G, Buratti C, Feltrin L, et al. (2018) Smart City Pilot Projects Using LoRa and IEEE802.15.4 Technologies. Sensors 18: 1118. [5]

Aoudia F, Gautier M, Magno M, et al. (2018) Long-short range communication network leveraging LoRa™ and wake-up receiver. Microprocessors and Microsystems 56: 184–192.[6]

Abhishek IJ ISE, The Oxford College of Engineering, Bengaluru, Karnataka, India
“LoRa-Based Emergency Communication Device Without Network or External Power” [7]